

OS-Product-8 Principles Schemes

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ENGINEERING MILESTONES

Key Project Achievements

- **Zero Black-Box Reliance:** Handcrafted Galois Field polynomial division in Python arrays without zlib.
- **Session Invariance:** Solved generator mismatch via dynamic 33-bit session polynomial singleton.
- **Production Delivery:** Fully documented codebase, Git repo, and cross-platform responsive dashboard.

RAPID-FIRE VIVA DEFENSE

Authoritative Answers

- **Q: Why Modulo-2?** Addition & subtraction are XOR without carries/borrows, enabling deterministic bitstream division.
- **Q: Why 33-bit divisor for 32-bit CRC?** Dividing by degree 32 guarantees remainder of degree ≤ 31 (exactly 32 bits).
- **Repository:** github.com/rogeralukka/OS-CN-Product-8